

evolution, it searches the available agent combination space. For an evolved agent aggregation (application) that is executed by the user, it then migrates to other peers (habitats) becoming hosted where it is useful, to combine with other agents in other populations to assist in responding to other user requests for applications.

6 Simulation and Results

We have previously compared the performance of Digital Ecosystems to traditional SOA based systems, and found that it outperforms at greater scales [6], and so here we focus on the self-organising behaviour which makes this possible. Self-organisation is perhaps one of the most desirable features in the systems that we design, and a primary motivation for our research in Digital Ecosystems is the desire to exploit the self-organising properties of biological ecosystems [25], which are thought to be robust, scalable architectures that can automatically solve complex, dynamic problems. Over time a biological ecosystem becomes increasingly self-organised through the process of *ecological succession* [3], driven by the evolutionary self-organisation of the populations within the ecosystem. Analogously, a Digital Ecosystem’s increasing self-organisation comes from the agent populations within being evolved to meet the dynamic *selection pressures* created by the requests from the user base. The self-organisation of biological ecosystems is often defined in terms of the *complexity*, *stability*, and *diversity* [19], which we will also apply to our Digital Ecosystems, here considering the *diversity*.

Self-organisation has been around since the late 1940s [1], but has escaped general formalisation despite many attempts [35, 20]. There have instead been many notions and definitions of self-organisation, useful within their different contexts [17]. They have come from cybernetics [1, 2, 18], thermodynamics [35], mathematics [24], information theory [41], synergetics [16], and other domains [23]. The term *self-organising* is widely used, but there is no generally accepted meaning, as the abundance of definitions would suggest. Therefore, the philosophy of self-organisation is complicated, because organisation has different meanings to different people. So, we would argue that any definition of self-organisation is context dependent, in the same way that a choice of statistical measure is dependent on the data being analysed.

While we could measure the self-organised *diversity* of individual evolving agent populations, or even take a random sampling, it will be more informative to consider their *collective* self-organised *diversity*. Also, given that the Digital Ecosystem is required to support a range of user behaviour, we can consider the *collective* self-organised *diversity* of the evolving agent populations relative to the *global* user request behaviour. So, when varying a behavioural property of the user requests according to some distribution, we would expect the corresponding property of the evolving agent populations to follow the same distribution. We are not intending to prescribe the expected user behaviour of the Digital Ecosystem, but investigate whether the Digital Ecosystem can adapt to a range of user behaviour. So, we will consider Uniform, Gaussian (Normal) and Power Law distributions for the parameters of the user request behaviour. The Uniform distribution will provide a control, while the Normal (Gaussian) distribution will provide a reasonable assumption for the behaviour of a large group of users, and the Power Law distribution will provide a relatively extreme variation in user behaviour. So, we simulated the Digital Ecosystem, varying aspects of the user behaviour according to different distributions, and measuring the related aspects of the evolving agent populations. This consisted of a mechanism to vary the user request properties of *length* and *modularity* (number of attributes per atomic service), according to

Uniform, Gaussian (normal) and Power Law distributions, and a mechanism to measure the corresponding application (agent aggregation) properties of *size* and *number of attributes per agent*. For statistical significance each scenario (experiment) will be averaged from ten thousand simulation runs. We expect it will be obvious whether the *observed* behaviour of the Digital Ecosystem matches the *expected* behaviour from the user base. Nevertheless, we will also implement a chi-squared (χ^2) test to determine if the observed behaviour (distribution) of the agent aggregation properties matches the expected behaviour (distribution) from the user request properties.

6.1 User Request Length

We started by varying the *user request length* according to the available distributions, expecting the size of the corresponding applications (agent aggregations) to be distributed according to the length of the user requests, i.e. the longer the user request, the larger the agent aggregation needed to fulfil it.

We first applied the Uniform distribution as a control, and graphed the results in Figure 6. The *observed* frequencies of the application (agent aggregation) size mostly matched the *expected* frequencies, which was *confirmed* by a χ^2 test; with a *null hypothesis* of *no significant difference* and *sixteen degrees of freedom*, the χ^2 value was 2.588, below the critical 0.95 χ^2 value of 7.962.

We then applied the Gaussian distribution as a reasonable assumption for the behaviour of a large group of users, and graphed the results in Figure 7. The *observed* frequencies of the application (agent aggregation) size matched the *expected* frequencies with only minor variations, which was confirmed by a χ^2 test; with a *null hypothesis* of *no significant difference* and *sixteen degrees of freedom*, the χ^2 value was 2.102, below the critical 0.95 χ^2 value of 7.962.

Finally, we applied the Power Law distribution to represent a relatively extreme variation in user behaviour, and graphed the results in Figure 8. The *observed* frequencies of the application (agent aggregation) size matched the *expected* frequencies with some variation, which was confirmed by a χ^2 test; with a *null hypothesis* of *no significant difference* and *sixteen degrees of freedom*, the χ^2 value was 5.048, below the critical 0.95 χ^2 value of 7.962.

There were a couple of minor discrepancies, similar to all the experiments. First, there were a small number of *individual* agents at the thousandth time step, caused by the typical user behaviour of continuously creating new services (agents). Second, while the chi-squared tests confirmed that there was no significant difference between the *observed* and *expected* frequencies of the application (agent aggregation) size, there was still a *bias* to larger applications (solutions). Evident visually in the graphs of the experiments, and evident numerically in the chi-squared test of the Power Law distribution experiment as it favoured shorter agent-sequences. The cause of this *bias* was most likely some aspect of *bloat* not fully controlled.

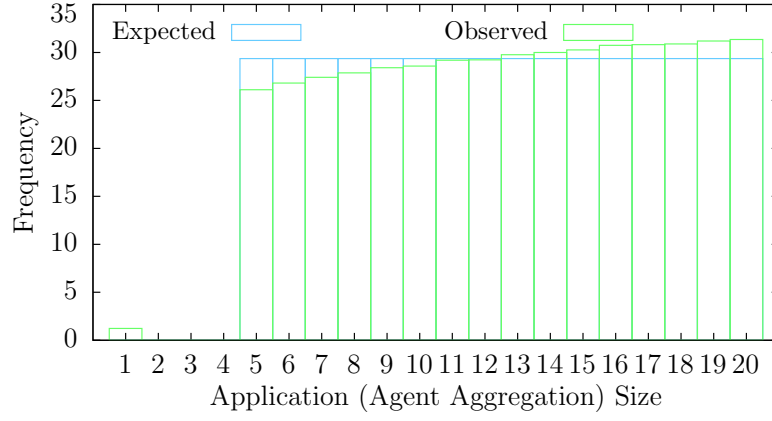


Figure 6: *Graph of Uniformly Distributed Agent-Sequence Length Frequencies: The observed frequencies of the application (agent aggregation) size mostly matched the expected frequencies.*

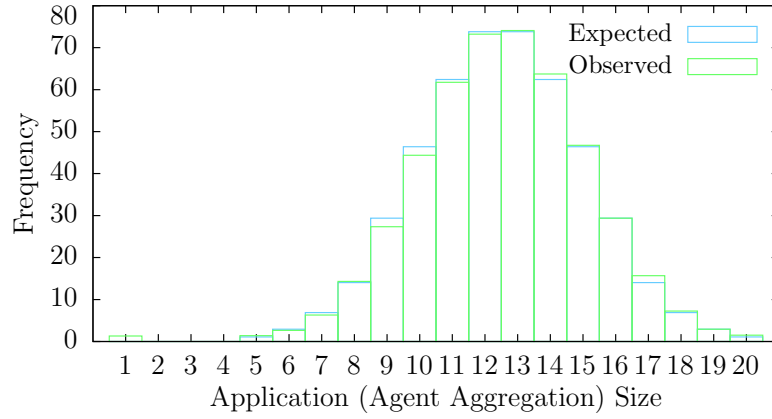


Figure 7: *Graph of Gaussian Distributed Agent-Sequence Length Frequencies: The observed frequencies of the application (agent aggregation) size matched the expected frequencies with only minor variations.*

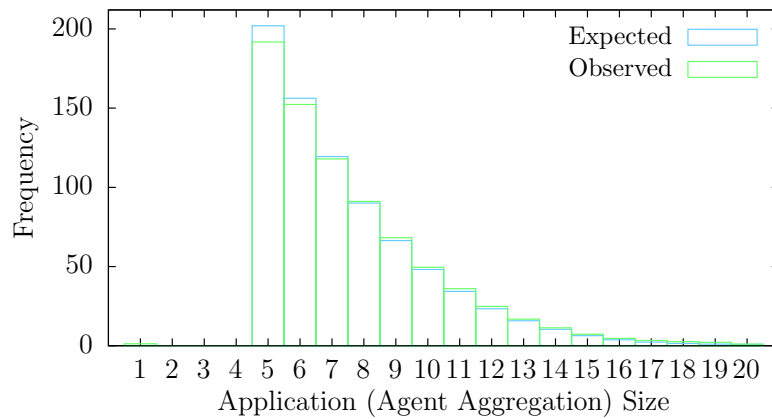


Figure 8: *Graph of Power Law Distributed Agent-Sequence Length Frequencies: The observed frequencies of the application (agent aggregation) size matched the expected frequencies with some variation.*